

Highlights

Machine Learning Models for Mental-Health Prediction from Actigraphy: A Systematic Appraisal and an Empirical Demonstration

- PROBAST+AI/TRIPOD+AI appraisal of 28 published models on open actigraphy cohorts
- Day-level cross-validation inflated AUROC by up to 0.17 and created signal from none
- Frozen cross-cohort transfer lost up to 0.21 AUROC with collapse of calibration
- No published model reports calibration; honest slopes ranged from 0.53 to 0.95
- Subject-wise, calibrated baselines released with public code and run manifests